# Mismatched Petz recovery for the Leung code under weak amplitude damping

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**Abstract**

The Petz recovery map is built from a model of the noise. What happens when this model is wrong? We study the four-qubit Leung code under amplitude damping with parameter $\gamma$, recovered by the Petz map built for an assumed parameter $h$. Numerically, the best choice is not $h = \gamma$ but a smaller value, which approaches $\gamma/2$ at weak noise. We prove that for $h = r\gamma$ with $r > 0$ fixed,

$$F(\gamma, r\gamma) = 1 - \left[\tfrac{7}{4} + \left(1 - \tfrac{1}{\sqrt{2}}\right) r(r-1)\right]\gamma^2 + O(\gamma^3),$$

so $r = \frac{1}{2}$ is optimal, and any underestimate of the noise ($0 < r < 1$) beats the matched map.

## 1 Introduction

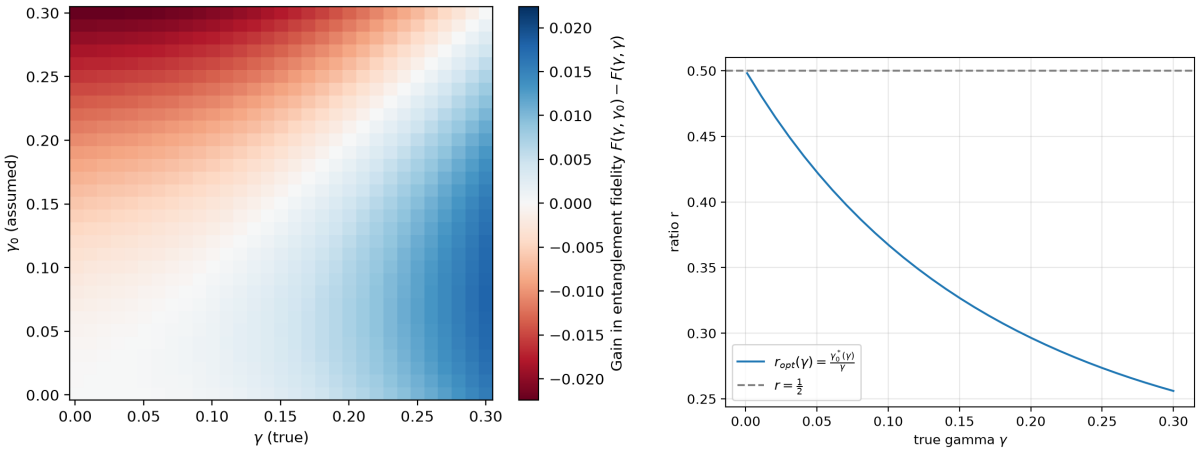
The Petz map is a universal recovery channel. Given a noise channel and a reference state, it provides an explicit decoder whose fidelity is provably close to the optimal one [1]. For the Leung code under amplitude damping, our SDP computations show that it is in fact much closer than this bound guarantees: $1-F_{\text{Petz}} \approx 1.75\gamma^2$, against $1-F_{\text{opt}} \approx 1.25\gamma^2$ for the optimal recovery [3]. Its weak point is that it needs the noise channel itself. In practice a damping rate is estimated, not known, so the decoder is built for an assumed parameter $h$ while the true parameter is $\gamma$. How much fidelity does a wrong guess cost?

We study this question for the four-qubit code of Leung, Nielsen, Chuang and Yamamoto [2] under independent amplitude damping. This setting was also investigated by Burri [4], who compared the performance of Petz recovery with other decoding strategies, including SDP-optimal recovery. Figure 1 illustrates a counterintuitive observation: a Petz recovery map constructed for an underestimated damping parameter can outperform the map constructed for the true noise. Figure 1(a) shows that the fidelity gain is positive below the diagonal $h = \gamma$ and negative above it, while Figure 1(b) shows that the optimal assumed parameter $h^*(\gamma)$ is systematically smaller than $\gamma$. In particular, the ratio $h^*(\gamma)/\gamma$ approaches $1/2$ in the weak-noise limit. This note provides an analytical explanation of this phenomenon in the weak-noise regime.

## 2 Mathematical setting

**Noise.** Amplitude damping with parameter $\gamma \in [0, 1]$ has Kraus operators $K_0(\gamma) = |0\rangle\langle 0| + \sqrt{1-\gamma}\,|1\rangle\langle 1|$ and $K_1(\gamma) = \sqrt{\gamma}\,|0\rangle\langle 1|$. On four qubits the Kraus operators are $N_j(\gamma) = K_{j_1}(\gamma) \otimes \cdots \otimes K_{j_4}(\gamma)$, labelled by bit strings $j = j_1 j_2 j_3 j_4$, where $j_i = 1$ means that qubit $i$ decayed. We write $|j|$ for the number of decays and $\bar{j}$ for the complementary string.

**Code.** The Leung code is spanned by $|0_L\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle)$ and $|1_L\rangle = \frac{1}{\sqrt{2}}(|0011\rangle + |1100\rangle)$. Its encoding isometry is $V = |0_L\rangle\langle 0| + |1_L\rangle\langle 1|$ and its code projector is $\Pi = VV^\dagger$. The code is supported on $\mathcal{C} = \mathcal{C}_0 \cup \mathcal{C}_1$, where $\mathcal{C}_0 = \{0000, 1111\}$ and $\mathcal{C}_1 = \{0011, 1100\}$. The encoded noise channel $\Phi_\gamma$ has the 16 Kraus operators $E_j(\gamma) = N_j(\gamma)V \in \mathbb{C}^{16\times 2}$.




(a) gain $F(\gamma, h) - F(\gamma, \gamma)$

(b) optimal ratio $h^*(\gamma)/\gamma$

Figure 1: Numerics from the notebooks `robustness.ipynb` and `petz_calibration.ipynb`; the assumed parameter $h$ is written $\gamma_0$ on the axes. (a) Gain of the Petz map built for $h$ over the matched map; blue means that the miscalibrated map is better. (b) The assumed parameter $h^*(\gamma)$ that maximises $F(\gamma, h)$, divided by $\gamma$. The dashed line is the weak-noise limit $\frac{1}{2}$ predicted by Theorem 1.

**Mismatched Petz map.** The Petz map built for the assumed parameter $h$, with reference state $I_2/2$, has the Kraus operators

$$R_k(h) = \frac{1}{\sqrt{2}}\, E_k(h)^\dagger\, \tau_h^{-1/2}, \qquad \tau_h = \Phi_h\Big(\frac{I_2}{2}\Big) = \frac{1}{2}\sum_j E_j(h) E_j(h)^\dagger. \tag{1}$$

For $0 < h < 1$, $\tau_h$ is invertible (Section 4). Hence $\sum_k R_k^\dagger R_k = \tau_h^{-1/2}\tau_h\tau_h^{-1/2} = I_{16}$, and the map is trace preserving without any completion. At $h = 0$, by contrast, $\tau_0 = \Pi/2$ is singular. With a pseudo-inverse and a completion on $\ker\tau_0$, as in the code, the map becomes the baseline recovery $\mathcal{R}_0(X) = V^\dagger X V + \mathrm{Tr}[(I-\Pi)X]\, I_2/2$.

**Fidelity.** The entanglement fidelity of a qubit channel with Kraus operators $L_i$ is $\frac{1}{4}\sum_i |\,\mathrm{Tr}\, L_i|^2$. The composition $\mathcal{R}_h \circ \Phi_\gamma$ has the 256 Kraus operators $R_k(h)E_j(\gamma)$, so

$$F(\gamma, h) = \frac{1}{8}\sum_{k,j} T_{kj}^2, \qquad T_{kj}(\gamma, h) = \mathrm{Tr}\left(E_k(h)^\dagger\, \tau_h^{-1/2}\, E_j(\gamma)\right) \in \mathbb{R}. \tag{2}$$

## 3 Main result

**Theorem 1.** *Fix $r > 0$ and set $h = r\gamma$. As $\gamma \to 0^+$,*

$$\boxed{F(\gamma, r\gamma) = 1 - c(r)\,\gamma^2 + O(\gamma^3), \qquad c(r) = \frac{7}{4} + \Big(1 - \frac{1}{\sqrt{2}}\Big)\, r(r-1).} \tag{3}$$

Since $r(r-1) = (r-\frac{1}{2})^2 - \frac{1}{4}$, the coefficient can also be written $c(r) = \frac{3}{2} + \frac{\sqrt{2}}{8} + (1-\frac{1}{\sqrt{2}})\,(r-\frac{1}{2})^2$. This form gives the consequences for calibration.

**Corollary 2.** *Let $c$ be as in (3).*

*(a)* Matched map: $1 - F(\gamma, \gamma) = \frac{7}{4}\gamma^2 + O(\gamma^3)$.

*(b)* Best calibration: *among Petz recoveries calibrated with $h = r\gamma$ for fixed $r > 0$, the leading-order infidelity is minimised at $r = \frac{1}{2}$, with $c(\frac{1}{2}) = \frac{3}{2} + \frac{\sqrt{2}}{8} \approx 1.677$. The gain of this calibration over the matched map is $F(\gamma, \frac{\gamma}{2}) - F(\gamma, \gamma) = \frac{2-\sqrt{2}}{8}\gamma^2 + O(\gamma^3) \approx 0.073\,\gamma^2$.*

*(c)* Sign of the error: *$c(r) < c(1)$ if and only if $0 < r < 1$. At leading order, any underestimate of the noise helps and any overestimate hurts, which is the sign pattern of Figure 1(a).*

*(d)* Optimal ratio: *if $0 < r_1 < \frac{1}{2} < r_2$, every maximiser $r^*(\gamma)$ of $r \mapsto F(\gamma, r\gamma)$ on $[r_1, r_2]$ satisfies $|r^*(\gamma) - \frac{1}{2}| = O(\sqrt{\gamma})$.*

*Proof.* Parts (a)–(c) follow from the two forms of $c$. For (d), let $f_\gamma(r) = (1 - F(\gamma, r\gamma))/\gamma^2$. The uniform remainder gives $|f_\gamma(r) - c(r)| \le C\gamma$ on $[r_1, r_2]$. Since $r^*$ minimises $f_\gamma$, we get $(1 - \frac{1}{\sqrt{2}})(r^* - \frac{1}{2})^2 = c(r^*) - c(\frac{1}{2}) \le f_\gamma(r^*) - f_\gamma(\frac{1}{2}) + 2C\gamma \le 2C\gamma$. □

Note also that $\lim_{r\to 0^+} c(r) = c(1) = \frac{7}{4}$. At this order, a Petz map built for a vanishingly small but non-zero noise is exactly as good as the matched one.

## 4 Proof outline

The proof is a direct computation, organised so that most of the 256 traces in (2) never need to be computed. The detailed matrices are collected in the appendix.

### Step 1: each Kraus column is a single basis vector, except for no decay

A decay pattern $j$ acts on a basis state by turning the decayed 1's into 0's:

$$N_j(\gamma)|x\rangle = \gamma^{|j|/2}(1-\gamma)^{(|x|-|j|)/2}|x - j\rangle \quad \text{if } j \preceq x, \qquad N_j(\gamma)|x\rangle = 0 \quad \text{otherwise,}$$

where $j \preceq x$ means that $x$ has a 1 wherever $j$ does. Applied to the codewords, this gives the two columns of $E_j$:

$$\sqrt{2}\,E_j(\gamma)|0\rangle = \delta_{j,0000}|0000\rangle + \gamma^{|j|/2}(1-\gamma)^{(4-|j|)/2}|\bar{j}\rangle, \tag{4}$$

$$\sqrt{2}\,E_j(\gamma)|1\rangle = \gamma^{|j|/2}(1-\gamma)^{(2-|j|)/2}(\mathbf{1}[j \preceq 0011]|0011 - j\rangle + \mathbf{1}[j \preceq 1100]|1100 - j\rangle). \tag{5}$$

Table 4 in the appendix lists all 32 columns. Two features matter:

- Every column is a multiple of a *single* basis vector, except the two columns of the no-decay operator $E_{0000}$, which lie in $\operatorname{span}\mathcal{C}_0$ and $\operatorname{span}\mathcal{C}_1$.
- A single decay ($|j| = 1$) sends $|0_L\rangle$ to $|\bar{j}\rangle$, of weight 3, and $|1_L\rangle$ to $|j'\rangle$, the other weight-one string of the same pair $\{0001, 0010\}$ or $\{0100, 1000\}$. The four single decays therefore leave eight distinct, mutually orthogonal syndromes.

### Step 2: $\tau_h$ is block diagonal

By (1), $\langle x|\tau_h|y\rangle = \frac{1}{2}\sum_j \left(N_j(h)^\dagger|x\rangle\right)^\dagger \Pi \left(N_j(h)^\dagger|y\rangle\right)$, and $N_j^\dagger|x\rangle \propto |x + j\rangle$ re-excites the decayed qubits. This vector is zero unless $x$ has a 0 wherever $j$ has a 1. Then each term in the sum is proportional to $\langle x + j|\Pi|y + j\rangle$, which is non-zero only when both strings lie in the same $\mathcal{C}_0$ or $\mathcal{C}_1$. For $x \neq y$ this forces $x + j$ and $y + j$ to be complementary. That is only possible if $j = 0000$ and $y = \bar{x}$ with $\{x, \bar{x}\} = \mathcal{C}_0$ or $\mathcal{C}_1$. Hence, after reordering the basis,

$$\tau_h \cong B_0(h) \oplus B_1(h) \oplus \lambda_3(h) I_4 \oplus \lambda_1(h) I_4 \oplus \lambda_2(h) I_4. \tag{6}$$

Here $B_0$ acts on $\mathcal{H}_0 = \mathrm{span}\{|0000\rangle, |1111\rangle\}$, $B_1$ acts on $\mathcal{H}_1 = \mathrm{span}\{|0011\rangle, |1100\rangle\}$, and the twelve remaining basis states are eigenvectors:

$$B_0 = \frac{1}{4}\begin{pmatrix} (1+h^2)^2 & (1-h)^2 \\ (1-h)^2 & (1-h)^4 \end{pmatrix}, \qquad B_1 = \frac{(1-h)^2}{4}\begin{pmatrix} 1+h^2 & 1 \\ 1 & 1+h^2 \end{pmatrix},$$

$\lambda_1 = \frac{1}{4}h(1-h)(1+h^2)$ $(|x|=1)$, $\lambda_2 = \frac{1}{4}h^2(1-h)^2$ $(x \in \mathcal{W})$, $\lambda_3 = \frac{1}{4}h(1-h)^3$ $(|x|=3)$,

where $\mathcal{W} = \{0101, 0110, 1001, 1010\}$ are the four weight-two strings outside $\mathcal{C}$. All blocks are positive definite for $0 < h < 1$, so $\tau_h$ is invertible. The inverse square root $\tau_h^{-1/2}$ has the same block structure, with explicit blocks given in Appendix B.

### Step 3: why only 20 of the 256 traces survive

Write $\tilde{e}_{j,\ell} = \sqrt{2}\, E_j|\ell\rangle$ for the columns. Then,

$$T_{kj} = \frac{1}{2}\sum_{\ell\in\{0,1\}} \tilde{e}_{k,\ell}(h)^T\, \tau_h^{-1/2}\, \tilde{e}_{j,\ell}(\gamma). \tag{7}$$

Two facts now decide which traces can be non-zero:

1. the trace only pairs *the same* column $\ell$ of $E_k(h)$ and $E_j(\gamma)$;
2. by (6), $\tau_h^{-1/2}$ maps each of the 14 orthogonal subspaces $\mathcal{H}_0$, $\mathcal{H}_1$ and $\mathbb{C}|x\rangle$ $(x \notin \mathcal{C})$ into themselves.

So $T_{kj} \neq 0$ requires some $\ell$ for which column $\ell$ of $E_k$ and column $\ell$ of $E_j$ lie in the same subspace. By Step 1, every non-zero column lies in exactly one subspace, as listed in Table 1.

| block | column $\ell = 0$ of $E_j$ for | column $\ell = 1$ of $E_j$ for |
|---|---|---|
| $\mathcal{H}_0$ | $j = 0000,\ 1111$ | $j = 0011,\ 1100$ |
| $\mathcal{H}_1$ | $j = 0011,\ 1100$ | $j = 0000$ |
| $\mathbb{C}\|x\rangle$, $\|x\| = 3$ | $j = \bar{x}$ | — |
| $\mathbb{C}\|x\rangle$, $\|x\| = 1$ | $j = \bar{x}$ | $j = x'$ |
| $\mathbb{C}\|x\rangle$, $x \in \mathcal{W}$ | $j = \bar{x}$ | — |

Table 1: Which columns land in which block of $\tau_h^{-1/2}$. The column $\ell = 1$ vanishes for $j \in \mathcal{W}$, $|j| = 3$ and $j = 1111$.

Reading the table:

- a block $\mathbb{C}|x\rangle$ receives at most one column for each $\ell$, so it only produces diagonal traces $T_{jj}$; this gives the $3 \times 4 = 12$ traces with $j \notin \mathcal{C}$;
- $\mathcal{H}_0$ with $\ell = 0$ couples 0000 and 1111, which gives 4 traces;
- $\mathcal{H}_1$ with $\ell = 0$ and $\mathcal{H}_0$ with $\ell = 1$ couple 0011 and 1100, which gives 4 traces;
- a mixed pair such as $(0000, 0011)$ never meets: for $\ell = 0$ its columns lie in $\mathcal{H}_0$ and $\mathcal{H}_1$, and for $\ell = 1$ in $\mathcal{H}_1$ and $\mathcal{H}_0$.

In total $12 + 4 + 4 = 20$ traces survive, and the other 236 vanish identically.

### Step 4: the twenty traces

Each surviving trace is a product of a few coefficients from (4)–(5) with one entry of $\tau_h^{-1/2}$. For a single decay ($|j| = 1$) the two columns lie in the blocks $\mathbb{C}|\bar{j}\rangle$ and $\mathbb{C}|j'\rangle$, where $\tau_h^{-1/2}$ is multiplication by $\lambda_3(h)^{-1/2}$ and by $\lambda_1(h)^{-1/2}$. This gives:

$$\begin{aligned} &\text{for } \ell = 0: \quad \tfrac{1}{2}\, h^{1/2}(1-h)^{3/2}\, \lambda_3(h)^{-1/2} = 1, \\ &\text{for } \ell = 1: \quad \tfrac{1}{2}\, \big(h(1-h)\big)^{1/2}\, \lambda_1(h)^{-1/2} = (1+h^2)^{-1/2}. \end{aligned} \tag{8}$$

Therefore

$$T_{(1)} := T_{jj} = \sqrt{\gamma(1-\gamma)}\,\Big[(1-\gamma) + \frac{1}{\sqrt{1+h^2}}\Big] \qquad (|j| = 1). \tag{9}$$

The other diagonal traces are $T_{(\mathcal{W})} = \gamma(1-\gamma)$ for $j \in \mathcal{W}$ and $T_{(3)} = h\gamma^{3/2}(1-\gamma)^{1/2}(1+h^2)^{-1/2}$ for $|j| = 3$. The no-decay trace involves both $B_0^{-1/2}$ and $B_1^{-1/2}$:

$$T_{0000,0000} = \frac{1 + (1-\gamma)^2 u}{t} + \frac{2(1-\gamma)}{w}, \tag{10}$$

where $w = \sqrt{2+h^2}$, $u = (1-h)^2 + hw$ and $t = \sqrt{1+u^2}$. The seven remaining traces, in $\mathcal{C}_0 \times \mathcal{C}_0$ and $\mathcal{C}_1 \times \mathcal{C}_1$, are listed in Appendix C. At order $\gamma^2$, only the pair $T_{0011,0011} = T_{1100,1100} = \gamma + O(\gamma^2)$ matters. With multiplicities, (2) becomes

$$8F(\gamma, h) = T_{0000,0000}^2 + 4T_{(1)}^2 + 4T_{(\mathcal{W})}^2 + 2T_{0011,0011}^2 + \rho(\gamma, h), \tag{11}$$

where $\rho$ collects the squares of the nine other traces (Appendix C). It is of order $\gamma^4$ when $h = r\gamma$.

### Step 5: why the first-order terms cancel

Put $h = r\gamma$ and expand. From (9) and (10),

$$T_{0000,0000}^2 = 8 - 16\gamma + O(\gamma^2), \qquad 4T_{(1)}^2 = 16\gamma + O(\gamma^2),$$

and all other terms of (11) are $O(\gamma^2)$. The two first-order terms cancel exactly, for every $r > 0$.

### Step 6: where the coefficient $c(r)$ comes from

At order $\gamma^2$, the $r$-dependence comes only from $T_{0000,0000}$. Up to $O(h^3)$, the quantities in (10) expand as

$$\frac{1}{t} = \frac{1}{\sqrt{2}}\Big[1 + \frac{2-\sqrt{2}}{2}\, h + \frac{1-2\sqrt{2}}{4}\, h^2\Big], \qquad \frac{u}{t} = \frac{1}{\sqrt{2}}\Big[1 - \frac{2-\sqrt{2}}{2}\, h + \frac{6\sqrt{2}-7}{4}\, h^2\Big],$$

$$\frac{2}{w} = \sqrt{2} - \frac{\sqrt{2}}{4}\, h^2.$$

We then split the first term of (10) as

$$\frac{1 + (1-\gamma)^2 u}{t} = \Big(\frac{1}{t} + \frac{u}{t}\Big) - (2\gamma - \gamma^2)\,\frac{u}{t},$$

where the terms linear in $h$ cancel inside the bracket. This gives

$$T_{0000,0000} = 2\sqrt{2} - 2\sqrt{2}\,\gamma + \Big[\frac{\sqrt{2}}{2} + \underbrace{(\sqrt{2}-1)\, r}_{\text{cross term } \gamma\cdot h} \; \underbrace{-\,(\sqrt{2}-1)\, r^2}_{\text{terms in } h^2}\Big]\gamma^2 + O(\gamma^3). \tag{12}$$

Squaring (12) and collecting all contributions to (11) gives Table 2.

Dividing by 8, and using $\frac{8-4\sqrt{2}}{8} = 1 - \frac{1}{\sqrt{2}}$, gives Theorem 1. As for the remainder: the factors $\sqrt{\gamma}$ in $T_{(1)}$ and $T_{(3)}$ only enter through their squares, so the right-hand side of (11) is a smooth function of $(\gamma, r)$ near $\gamma = 0$. Taylor's theorem then bounds the remainder by $O(\gamma^3)$, uniformly for $r$ in compact sets.

| contribution | $\gamma^0$ | $\gamma^1$ | $\gamma^2$ |
|---|---|---|---|
| $T^2_{0000,0000}$ | 8 | $-16$ | $12 + (8 - 4\sqrt{2})\, r(1-r)$ |
| $4T^2_{(1)}$ | 0 | $+16$ | $-32$ |
| $4T^2_{(\mathcal{W})}$ | 0 | 0 | $+4$ |
| $2T^2_{0011,0011}$ | 0 | 0 | $+2$ |
| $\rho$ | 0 | 0 | 0 |
| $8F(\gamma, r\gamma)$ | 8 | 0 | $-14 + (8 - 4\sqrt{2})\, r(1-r)$ |

Table 2: Coefficients of $\gamma^0$, $\gamma^1$ and $\gamma^2$ in (11) at $h = r\gamma$.

## 5 Numerical verification and discussion

**Leading coefficient.** Table 3 compares $(1 - F(\gamma, r\gamma))/\gamma^2$, computed with the function `mismatched_petz_fidelity`, with $c(r)$. The difference shrinks linearly in $\gamma$, as expected from the $O(\gamma^3)$ remainder. Below $\gamma \approx 10^{-4}$ the numerical code loses accuracy, because of the eigenvalues of order $h^2$ in $\tau_h$.

| $r$ | $\gamma = 10^{-1}$ | $\gamma = 10^{-2}$ | $\gamma = 10^{-3}$ | $c(r)$ |
|---|---|---|---|---|
| 1/4 | 1.6298 | 1.6889 | 1.6945 | 1.6951 |
| 1/2 | 1.6311 | 1.6728 | 1.6764 | 1.6768 |
| 1 | 1.7624 | 1.7528 | 1.7503 | 1.7500 |
| 2 | 2.5087 | 2.3580 | 2.3380 | 2.3358 |

Table 3: $(1 - F(\gamma, r\gamma))/\gamma^2$ from the code, compared with the coefficient $c(r)$ of Theorem 1.

**Optimal calibration.** The optimiser `optimal_petz_calibration` gives $h^*/\gamma = 0.497$ at $\gamma = 10^{-3}$ and 0.482 at $\gamma = 10^{-2}$, approaching $\frac{1}{2}$ (Figure 1(b)). At $\gamma = 10^{-3}$ the gain over the matched map is $7.39 \times 10^{-8}$, against $\frac{2-\sqrt{2}}{8}\gamma^2 = 7.32 \times 10^{-8}$. At larger $\gamma$ higher orders take over: the optimal ratio drops to 0.37 at $\gamma = 0.1$ and 0.26 at $\gamma = 0.3$.

**The singular case $h = 0$.** An interesting feature of the Petz recovery is its discontinuous behaviour at $h = 0$. For every fixed $r > 0$, choosing $h = r\gamma$ gives $1 - F(\gamma, r\gamma) = O(\gamma^2)$ as $\gamma \to 0^+$. In contrast, setting $h = 0$ reduces the completed Petz map to the baseline recovery, for which $1 - F(\gamma, 0) = \frac{3}{2}\gamma + O(\gamma^2)$. This distinction arises because $\tau_h$ is full rank for $h > 0$ but has rank two at $h = 0$.

**Comparison with optimal recovery.** The SDP-optimal recovery gives $(1 - F_{\text{opt}})/\gamma^2 \approx 1.250$ for $\gamma \leq 0.05$ (`data/fidelities.csv`). At weak noise, the best calibration therefore closes only $(c(1) - c(\frac{1}{2}))/(c(1) - 1.25) \approx 15\%$ of the gap between the matched Petz map and the optimum. It helps more at stronger noise: at $\gamma = 0.3$ it raises $F$ from 0.849 to 0.867, against $F_{\text{opt}} = 0.890$, which closes 44% of the gap. The constant 1.25 is a numerical observation, not part of the theorem.

**Practical message and limitations.** For sufficiently weak amplitude damping and a fixed relative calibration $h = r\gamma$, choosing $0 < r < 1$ improves the leading-order fidelity compared with matched Petz recovery, and the ratio $\frac{1}{2}$ is optimal. The guess should never be set to zero, though. Theorem 1 is asymptotic: it describes the regime $\gamma \to 0$. At finite $\gamma$ the higher orders matter: the optimal ratio drifts below $\frac{1}{2}$.

# A Kraus columns

Table 4 lists the columns of $\sqrt{2}\,E_j(\gamma)$ from (4)–(5), grouped by type of decay. For $|j| = 1$, $j'$ is the partner defined in Step 1.

| $j$ | $\sqrt{2}\,E_j(\gamma)\|0\rangle$ | $\sqrt{2}\,E_j(\gamma)\|1\rangle$ |
|---|---|---|
| 0000 | $\|0000\rangle + (1-\gamma)^2\|1111\rangle$ | $(1-\gamma)(\|0011\rangle + \|1100\rangle)$ |
| $\|j\| = 1$ | $\gamma^{1/2}(1-\gamma)^{3/2}\|\bar{j}\rangle$ | $\gamma^{1/2}(1-\gamma)^{1/2}\|j'\rangle$ |
| 0011, 1100 | $\gamma(1-\gamma)\|\bar{j}\rangle$ | $\gamma\|0000\rangle$ |
| $j \in \mathcal{W}$ | $\gamma(1-\gamma)\|\bar{j}\rangle$ | 0 |
| $\|j\| = 3$ | $\gamma^{3/2}(1-\gamma)^{1/2}\|\bar{j}\rangle$ | 0 |
| 1111 | $\gamma^2\|0000\rangle$ | 0 |

Table 4: The columns of $\sqrt{2}\,E_j(\gamma)$ in the computational basis.

# B The blocks of $\tau_h^{-1/2}$

Throughout, $w = \sqrt{2+h^2}$, $u = (1-h)^2 + hw$ and $t = \sqrt{1+u^2}$, as in (10).

**The $\mathcal{H}_1$ block.** $B_1$ has the eigenvectors $\frac{1}{\sqrt{2}}(1, \pm 1)$, with eigenvalues

$$\mu_+ = \tfrac{1}{4}(1-h)^2(2+h^2), \qquad \mu_- = \tfrac{1}{4}h^2(1-h)^2.$$

Hence

$$B_1^{-1/2} = \begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}, \qquad \alpha = \frac{1}{1-h}\Big(\frac{1}{h} + \frac{1}{w}\Big), \qquad \beta = \frac{1}{1-h}\Big(\frac{1}{w} - \frac{1}{h}\Big).$$

**The $\mathcal{H}_0$ block.** For a $2 \times 2$ positive definite matrix $M$, the Cayley–Hamilton theorem gives $\sqrt{M} = (M + \sqrt{\det M}\, I)/\sqrt{\operatorname{Tr} M + 2\sqrt{\det M}}$. Applied to $M = 4B_0$, for which $\sqrt{\det M} = (1-h)^2 hw =: s$ and $\operatorname{Tr} M + 2s = 1 + u^2 = t^2$, this gives

$$B_0^{-1/2} = \frac{2}{s\,t}\begin{pmatrix} (1-h)^4 + s & -(1-h)^2 \\ -(1-h)^2 & (1+h^2)^2 + s \end{pmatrix} = \begin{pmatrix} a_0 & c_0 \\ c_0 & d_0 \end{pmatrix},$$

with $a_0 = \dfrac{2u}{hwt}$, $c_0 = -\dfrac{2}{hwt}$ and $d_0 = \dfrac{2(1+hwu)}{(1-h)^2 hwt}$.

## C The remaining traces

Each trace follows from (7) and Table 1. For instance, the columns of $E_{0000}$ give, with the identity $u-(1-h)^2=hw$,

$$\begin{aligned} T_{0000,0000} &= \tfrac{1}{2}(1,(1-h)^2)B_0^{-1/2}(1,(1-\gamma)^2)^T + \tfrac{1}{2}(1-h)(1-\gamma)\,(1,1)\,B_1^{-1/2}(1,1)^T \\ &= \frac{1+(1-\gamma)^2u}{t} + \frac{2(1-\gamma)}{w}. \end{aligned}$$

In the same way,

$$T_{0000,1111} = \frac{\gamma^2}{t}, \qquad T_{1111,0000} = \frac{h(u-(1-\gamma)^2)}{wt}, \qquad T_{1111,1111} = \frac{h\gamma^2 u}{wt},$$

$$T_{0011,0011} = T_{1100,1100} = \frac{\gamma}{2}\Big[(1-\gamma)\Big(1+\frac{h}{w}\Big) + \frac{2u}{wt}\Big],$$

$$T_{0011,1100} = T_{1100,0011} = \frac{\gamma}{2}\Big[(1-\gamma)\Big(\frac{h}{w}-1\Big) + \frac{2u}{wt}\Big].$$

The remainder in (11) is

$$\rho = T_{0000,1111}^2 + T_{1111,0000}^2 + T_{1111,1111}^2 + 2T_{0011,1100}^2 + 4T_{(3)}^2.$$

At $h=r\gamma$ every term of $\rho$ is of order $\gamma^4$ or smaller, since $T_{0011,1100}$, $T_{0000,1111}$ and $T_{1111,0000}$ are $O(\gamma^2)$, $T_{1111,1111}=O(\gamma^3)$ and $T_{(3)}=O(\gamma^{5/2})$.

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An AI assistant (Claude, Anthropic) was used to assist with the extensive computations. The outline of the proof and research direction were developed by the author. The resulting mathematical identities were checked using symbolic and numerical computations.

## References

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